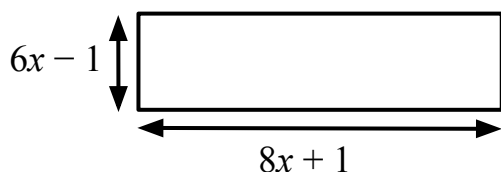




Problem solving with linear equations

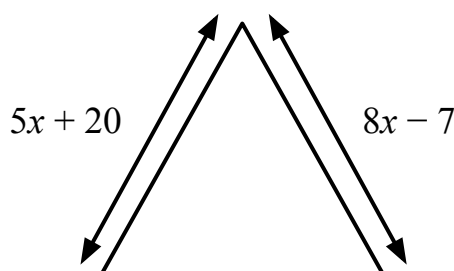
Task A: Solving geometric problems

1) The perimeter of this rectangle is 7. Work out the value of x



$$\begin{aligned} 2(6x - 1) + 2(8x + 1) &= 7 \\ 12x - 2 + 16x + 2 &= 7 \\ 28x &= 7 \\ x &= \frac{7}{28} = \frac{1}{4} \end{aligned}$$

2) Find the perimeter of this equilateral triangle.



An equilateral triangle has equal lengths.

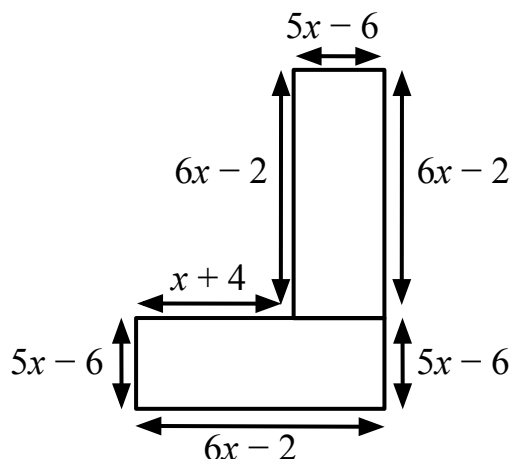
$$\begin{aligned} 5x + 20 &= 8x - 7 \\ 20 &= 3x - 7 \\ 27 &= 3x \\ x &= 9 \end{aligned}$$

$$5x + 20 = 5(9) + 20 = 45 + 20 = 65$$

The perimeter is $3 \times 65 = 195$

3) Two identical rectangles make this compound shape with perimeter 48.

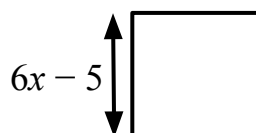
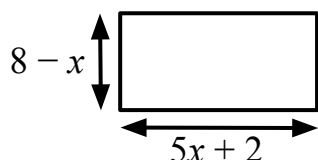
Work out the value of x



$$\begin{aligned} (6x - 2) - (x + 4) \\ &= 6x - 2 - x - 4 \\ &= 5x - 6 \end{aligned}$$

$$\begin{aligned} 3(6x - 2) + 3(5x - 6) + (x + 4) &= 48 \\ 18x - 6 + 15x - 18 + x + 4 &= 48 \\ 34x - 20 &= 48 \\ 34x &= 68 \\ x &= 2 \end{aligned}$$

4) The rectangle and the square have equal perimeters. Work out this perimeter.



$$\begin{aligned} 2(8 - x) + 2(5x + 2) &= 4(6x - 5) \\ 16 - 2x + 10x + 4 &= 24x - 20 \\ 8x + 20 &= 24x - 20 \\ 40 &= 16x \\ x &= 2.5 \end{aligned}$$

$$6x - 5 = 6(2.5) - 5 = 10$$

$$4 \times 10 = 40$$

The perimeter of both shapes is 40



Task B: Forming and solving equations from worded problems

1) a) An electrician has a call out fee and then charges £45 per hour. For an eight hour job they charge £385. Form an equation and calculate their call out fee.

$$c + 8 \times £45 = £385$$

$$c + £360 = £385$$

$$c = £25$$

Their call out fee is £25

b) Another electrician has a call out fee of £50 and for an eight hour job they charge £370. Form an equation to calculate their hourly rate.

$$£50 + 8x = £370$$

$$8x = £320$$

$$x = £40$$

Their hourly rate is £40

2) Andeep has made two number problems. Write an equation and find the number in each case.

a) I have a number whereby if you multiply it by 3 and add 60 it's the same as if you multiply it by 7 and minus 8

$$3x + 60 = 7x - 8$$

$$3x + 68 = 7x$$

$$68 = 4x$$

$$x = 17$$

Andeep's number is 17

b) I have another number whereby if you multiply it by 5 and minus 7 it's the same as if you multiply it by 2 and minus 5

$$5x - 7 = 2x - 5$$

$$3x - 7 = -5$$

$$3x = 2$$

$$x = \frac{2}{3}$$

Andeep's number is $\frac{2}{3}$

3) Jun, Aisha and Jacob are buying a present for classmates.

a) Aisha spends £2 more than Jun and Jacob spends three times as much as Jun. In total they spend £39.50. How much does Jacob spend?

Jun	x			
Aisha	x	2		
Jacob	x	x	x	

£39.50

$$5x + 2 = 39.50$$

$$5x = 37.50$$

$$x = 7.50$$

Jacob spends $3 \times £7.50 = £22.50$

b) Aisha spends £5 more than Jun and Jacob spends four times as much as Aisha. Jacob spends £33 more than Aisha. How much does Jun spend?

Jun	x						
Aisha	x	5					
Jacob	x	5	x	5	x	5	x

£33

$$3(x + 5) = 33$$

$$x + 5 = 11$$

$$x = 6$$

Jun spends £6

